

SANTIAGO G. MEDINA-MUÑOZ

PhD Candidate in Integrative Biology — Population Genetics & Computational Biology

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I study how demographic history—migration, admixture, and drift—has shaped genetic variation in human and non-human populations. My doctoral work combines whole-genome sequencing with coalescent-based demographic inference to reconstruct the pre-Columbian and post-contact history of Mexican populations. Earlier work examined how codon composition shapes mRNA stability and translation. I am looking for a postdoctoral position applying statistical and population genomic methods to questions in evolutionary demography, ancient DNA, or comparative genomics.

Education

PhD in Integrative Biology

CINVESTAV, LANGEBIO — Irapuato, Mexico

2022–Present

Topic: Evolutionary history of pre-Columbian Mexico. Advisors: Andrés Moreno-Estrada & Aaron Ragsdale.

MSc in Integrative Biology

CINVESTAV, LANGEBIO — Irapuato, Mexico

2020–2022

Topic: Demographic inference in admixed populations. Advisor: Andrés Moreno-Estrada.

BSc in Agro-genomic Sciences

Universidad Nacional Autónoma de México — León, Mexico

2013–2017

Publications

Preprints

Snodgrass S., Li F., Mambakkam S., **Medina-Muñoz S.G.**, . . . , Ross-Ibarra J. Maize genetic diversity is largely unstructured by human ethnolinguistic diversity in its center of origin. *bioRxiv*, 2026. doi:10.1101/2026.02.02.703385

Peer-reviewed journal articles

8. Barberena-Jonas C., **Medina-Muñoz S.G.**, et al. Clinical genetic variation across Hispanic populations in the Mexican Biobank. *Nature Medicine*, 32:725–735, 2026. doi:10.1038/s41591-025-04100-z
7. Sohail M., Palma-Martínez M.J., . . . , **Medina-Muñoz S.G.**, et al. Mexican Biobank advances population and medical genomics of diverse ancestries. *Nature*, 622(7984):775–783, 2023. doi:10.1038/s41586-023-06560-0
6. **Medina-Muñoz S.G.**, Ortega-Del Vecchyo D., Cruz-Hervert L.P., Ferreyra-Reyes L., García-García L., Moreno-Estrada A., Ragsdale A.P. Demographic modeling of admixed Latin American populations from whole genomes. *The American Journal of Human Genetics*, 110(10):1804–1816, 2023. doi:10.1016/j.ajhg.2023.08.015
5. *Diez M., ***Medina-Muñoz S.G.**, Castellano L.A., et al. iCodon customizes gene expression based on the codon composition. *Scientific Reports*, 12(1):12126, 2022. doi:10.1038/s41598-022-15526-7

4. Rodríguez-Rodríguez J.E., Ioannidis A.G., **Medina-Muñoz S.G.**, Barberena-Jonas C., Blanco-Portillo J., Quinto-Cortés C.D., Moreno-Estrada A. The genetic legacy of the Manila galleon trade in Mexico. *Philosophical Transactions of the Royal Society B*, 377(1852):20200419, 2022. doi:10.1098/rstb.2020.0419
3. Jiménez-Kaufmann A., Chong A.Y., . . . , **Medina-Muñoz S.G.**, et al. Imputation performance in Latin American populations: improving rare variant representation with the inclusion of Native American genomes. *Frontiers in Genetics*, 12:719791, 2022. doi:10.3389/fgene.2021.719791
2. **Medina-Muñoz S.G.**, Kushawah G., Castellano L.A., et al. Crosstalk between codon optimality and cis-regulatory elements dictates mRNA stability. *Genome Biology*, 22:1–23, 2021. doi:10.1186/s13059-020-02251-5
1. Wu Q., **Medina-Muñoz S.G.**, Kushawah G., et al. Translation affects mRNA stability in a codon-dependent manner in human cells. *eLife*, 8:e45396, 2019. doi:10.7554/eLife.45396

* equal contribution. Full list with citation metrics on [Google Scholar](#).

Research Experience

PhD Fellow — Computational Genomics & Population Genetics

CINVESTAV LANGE BIO, Irapuato, Mexico

2020–Present

Sup. A. Moreno-Estrada & A. Ragsdale

Reconstructing fine-scale demographic history of Indigenous and admixed Mexican lineages from whole-genome sequence data using coalescent and site-frequency-spectrum methods.

Honorary Fellow

University of Wisconsin–Madison, USA

Summer 2023

Sup. A. Ragsdale

Extended PhD project on demographic inference of Mexican populations using whole-genome data.

Bioinformatician

Stowers Institute, Kansas City, USA

2017–2020

PI A. Bazzini

Studied how gene regulatory information is encoded in the coding sequence through codon optimality.

Undergraduate Research Thesis

Stowers Institute, Kansas City, USA

2016–2017

PI A. Bazzini

Thesis: “The regulatory grammar of codon optimality.”

Stowers Summer Scholar

Stowers Institute, Kansas City, USA

Summer 2015

PI J. Zeitlinger

Developed a pipeline for analyzing protein-DNA binding interactions with ChIP-nexus data.

Undergraduate Summer Research

CINVESTAV LANGE BIO, Irapuato, Mexico

Summer 2014

PI C. Abreu-Goodgear

Investigated conservation of micro-RNA targets across vertebrate species using RNA-seq data.

Technical Skills

Population genetics: demographic inference (moments, msprime, ABC), coalescent simulation, site-frequency-spectrum modeling, admixture & local ancestry inference.

Bioinformatics: whole-genome variant calling (GATK, bcftools), read alignment, RNA-seq & ChIP-nexus analysis pipelines.

Programming & tools: Python, R, Bash, Snakemake, Git, HPC/Slurm cluster computing.

Funding, Fellowships & Awards

Graduate

- EMBO | EMBL Symposia Fellowship — *fully funded travel & participation, EMBO Symposium: Reconstructing the Human Past* (Oct 2024)
- 3rd Place, Best Oral Presentation — *6th Mexico Population Genomics Meeting, Cuernavaca* (Oct 2023)
- Young Investigator Travel Award — *SMBE, Ferrara, Italy* (July 2023)
- Elisa Acuña Scholarship — *paid internship, University of Wisconsin-Madison* (April 2023)
- CONACYT Scholarship — *stipend covering MSc and PhD studies, CINVESTAV* (Aug 2020)

Undergraduate

- Stowers Summer Scholar — *Stowers Institute, Kansas City* (Summer 2015)
- Colgate Academic Excellence — *ENES UNAM, support for research involvement* (2016)

Presentations

Invited talks

- Inferring fine-scale demography of Indigenous lineages from 500 whole genomes across Mexico — *SMBE24, Puerto Vallarta, Mexico* (July 2024)
- Demographic modeling of admixed Latin American populations from whole genomes — *6th Mexico Population Genomics Meeting, Cuernavaca-Morelos* (Oct 2023)
- Demographic modeling of admixed Latin American populations from whole genomes — *SMBE23, Ferrara, Italy* (July 2023)

Posters

- Inferring fine-scale demography of Indigenous lineages from 500 whole genomes across Mexico — *Reconstructing the Human Past*, EMBL Heidelberg (2024)
- Deciphering mRNA decay programs during embryogenesis — *Biological Data Science Meeting*, Cold Spring Harbor Laboratory (2018)
- A pipeline for peak calling and motif discovery in ChIP-nexus data — *XVI National Congress of Plant Biochemistry*, Querétaro (2016)

Workshops & courses

- Population genomics: background and tools — *EMBO courses (virtual)* (2022)
- NCBI Hackathon — *Cold Spring Harbor Laboratory*, New York (2018)
- Statistical Data Analysis for Genome Scale Biology — *EMBO, Bresanone, Italy* (2017)
- Bioinformatics Summer School — *Institute of Mathematics UNAM*, Querétaro (2015)

Teaching & Mentorship

Agrega aquí ayudantías, cursos impartidos o mentoría de estudiantes.

Professional Service

- Manuscript reviewer — *Genome Biology and Evolution* (GBE)

Agrega aquí membresías o participación en comités.

References

Andrés Moreno-Estrada

CINVESTAV LANGE BIO — *PhD advisor*

andres.moreno@cinvestav.mx

Aaron Ragsdale

University of Wisconsin–Madison — *PhD co-advisor*

apragdale@wisc.edu

Ariel Bazzini

Stowers Institute for Medical Research — *Research supervisor*

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